

Navigators IDR — Design Requirements

Smart India Hackathon 2026 (SIH26168) — ISRO Problem Statement

1. System Objective

The system provides intelligent dead reckoning (IDR) using smartphone-only sensors to estimate vehicle trajectory during GNSS outages (e.g., tunnels, forests, urban canyons).

2. Hard Requirements

- **Smartphone-Only:** Must not rely on OBD-II, CAN bus, vehicle speedometers, or dedicated FOG-IMUs.
- **Offline Capable:** The core inference and EKF must run without an internet connection, allowing the frontend (JavaScript) to estimate position seamlessly. The Python backend is explicitly designated as a **development and testing tool**.
- **No Ground-Truth Leakage:** The estimator must never have access to ground-truth velocity or position during an outage.

3. Core Architecture Components

1. **Sensor Interface:** Captures linear acceleration and gyroscope data at 100 Hz.
2. **Phone-to-Vehicle Alignment:** Dynamically aligns the arbitrary smartphone frame to the vehicle's Forward/Lateral/Vertical frame using gravity and forward acceleration vectors.
3. **Temporal Convolutional Network (TCN):** A lightweight, low-latency ONNX model that predicts a 2D planar velocity vector from a 200-timestep window of IMU data.
4. **15-State Extended Kalman Filter (EKF):** Fuses the AI velocity predictions with raw IMU data, and incorporates GNSS when available.
5. **Non-Holonomic Constraints (NHC):** Restricts lateral and vertical velocity.
6. **Zero Velocity Updates (ZUPT):** Freezes drift when the vehicle is stationary.

4. Benchmark Validation Status

Initial invalid benchmarks were flagged during forensic validation due to ground-truth leakage. The new benchmark suite cleanly firewalls ground truth. Current TCN validation metrics show ~90% drift (compared to 400%+ previously), proving the end-to-end integration is functional. Future work involves training the TCN on real-world datasets to bridge the gap to the <10% drift target.